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α-Glucosidase inhibitory activity of mangiferin-loaded F127/PEG micellar system

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ABSTRACT

Mangiferin-loaded F127/PEG micellar systems were evaluated regarding the effect of PEG addition at different molecular weights, where the system with best mangiferin solubility response (0.5 wt% of PEG in 1 wt% F127 aqueous solution) was chosen to be a potential drug delivery system (DDS), which solubilized 7.92 mg dL^{-1} of mangiferin. Subsequently, the chosen mangiferin-loaded DDS was evaluated in regard the inhibition of enzyme α-glucosidase, which showed a more effective inhibition, around 95.42%, when compared to the standard acarbose (92.59%) and to the free-form mangiferin (90.42%). The same micellar system was also evaluated for its cytotoxicity, where no significative values of LDH activity was observed. No study has been done yet with both mangiferin and DDSs regarding *in vitro* α-glucosidase inhibitory activity. This work proposes a "new" alternative towards diabetes treatment.

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1. Introduction

Diabetes mellitus, commonly known as Type 2 Diabetes, is an endocrine metabolic disorder. It is believed that the number of people with Type 2 Diabetes will reach more than 800 million worldwide by 2045 [1]. In this context, mangiferin has gained wide attention regarding its outstanding α-glucosidase inhibitory activity, as well as other characteristics such as anticancer and anti-inflammatory [1–4]. However, due to poor water-soluble limitation, mangiferin leads to a low bioavailability [5]. Therefore, new approaches related to "smart" DDSs are required to increase the bioavailability of hydrophobic drugs. Among these, self-assembly mixtures of block copolymers such as poly(ethyleneglycol) (PEG) and Pluronic ® F127 have been used to overcome the solubility issues associated with hydrophobic drugs [6,7]. To our knowledge, it is the first working dealing with mangiferin-loaded micellar system and its α-glucosidase inhibitory activity [5,8]. This work reports a mangiferin-loaded F127/PEG micellar solution, where all used components are FDA-approved [9]. In addition, with the purpose of future clinical trial tests, the mangiferin/F127/PEG micellar solution was also analyzed through cytotoxicity assay [3,10].

2. Experimental procedure

2.1. Materials

The chemical characterisation of PEG 400, 6,000 and 35,000 g mol^{-1}(denoted as PEG 400, 6 K and 35 K) and Pluronic® F127 were previously reported [6]. α-Glucosidase enzyme was purchased from Sigma-Aldrich. The positive control acarbose (Glucobay®) was produced by Bayer. Mangiferin (422.3 g mol^{-1}) was donated by NPL (UFC) and used as received. Human leucocyte-rich blood from healthy adults was donated by HEMOCE blood bank, Fortaleza-CE, Brazil. All other reagents were of analytical grade.

2.2. Methods

2.2.1. Solubilisation

A stock aqueous solution of F127 1 wt% was used to prepare further solutions containing PEG 400, 6 K and 35 K in the range of 0.1–10 wt%. Mangiferin (10 mg) was added into 10 mL of these solutions. The systems were subsequently stirred at 37 °C for 48 h. After solubilisation procedure, "shake-flask" method, the samples were filtered and diluted in methanol for further quantification by UV/Vis spectroscopy [11]. The quantification was determined through calibration curve ($R^2 = 0.99973$) of mangiferin

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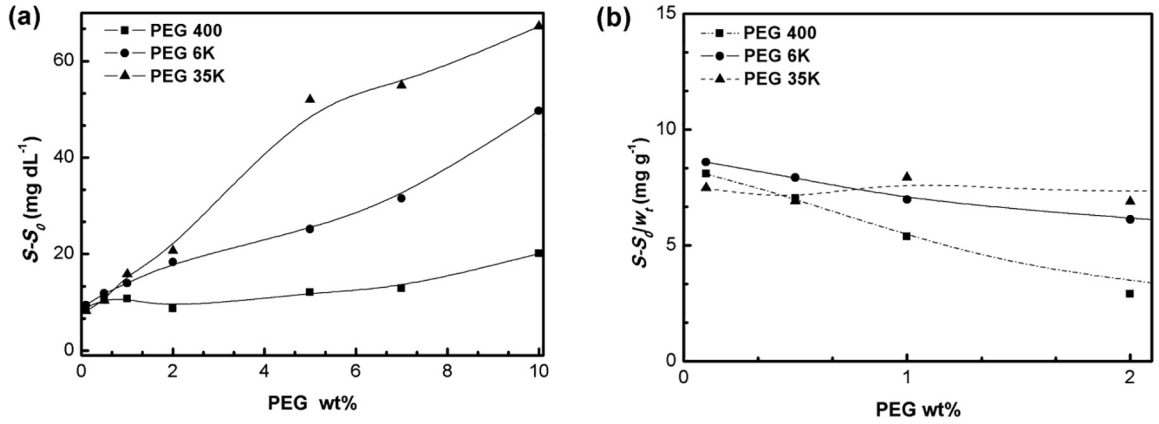


Fig. 1. Mangiferin solubility in micellar systems at 37 °C, considering S_0 = 18.1 mg dL^{-1}: (a) PEG addition evaluation at F127 1 wt% aqueous solution, without polymer weight consideration, and (b) PEG addition evaluation to 1 wt% of F127.

Table 1
α-Glucosidase inhibitory activity and IC_{50} values. Acarbose is used as positive control.

Samples	[C] (mg mL^{-1})[a]	Inhibition%	IC_{50} (mg mL^{-1})
Free-form mangiferin	1.0	90.42 ± 0.0002	0.283 ± 0.01
Mangiferin-loaded PEG/F127	1.49[b]	95.42 ± 3.75	0.128 ± 0.02
Acarbose	100.00	92.59 ± 0.09	4.26 ± 0.065

[a] [C] = Concentration of the sample at maximum inhibitory activity.
[b] Concentration of mangiferin loaded into the system.

(λ_{max} = 256 nm, Abs = 0.06011[*S*] – 0.0048, where *S* is the solubilisation capacity of the DDS). The same procedure was carried out for the free-form mangiferin in order to obtain drug solubility in water (S_0).

2.2.2. α-Glucosidase inhibitory activity

The enzymatic assay was adapted following the well-known α-glucosidase inhibitory activity method [3]. In this test, the system mangiferin/1 wt% F127/0.5 wt% PEG 6 K was chosen to act as a representative sample. Firstly, each sample was premixed with α-glucosidase 1 U mL^{-1} and incubated in phosphate buffer (pH 6.8) during 5 min at 25 °C. Then, the reaction was carried out after addition of *p*NPG 1 mM, which was further incubated during 30 min at 37 °C under stirring (Thermomixer comfort, Eppendorf®). The reaction was stopped by the addition of $NaHCO_3$ (10 wt%). Finally, the treated sample was added to a 96 well-plate, and the α-glucosidase activity was determined using a reader Elisa Thermoplate (405 nm) measuring the amount of *p*-nitrophenol released from *p*NPG hydrolysis reaction. Acarbose was used as a positive control.

2.2.3. Cytotoxicity

The lactate dehydrogenase (LDH) assay was performed similarly to a previous work [10], where the unloaded system (1wt% F127/0.5 wt% PEG 6 K) and both drug free-form and drug-loaded one were evaluated at 10, 50 and 100 μg mL^{-1}.

3. Results and discussion

Fig. 1a and b show the PEG addition influence in drug-loaded F127 micellar solution at 1 wt%. According to previous reported works, the solubilisation capacity of micellar systems is defined as $S–S_0/w_t$, mg g^{-1}, where *S* is the solubility of the drug in the solution, S_0 the solubility of drug in water, and w_t the total weight of polymers used [6,7].

In Fig. 1a, mangiferin solubility increases with PEG addition in 1 wt% F127 solution. The result is evidenced for concentrations of PEG 6 K and 35 K in the range of 2–10 wt% and even more for

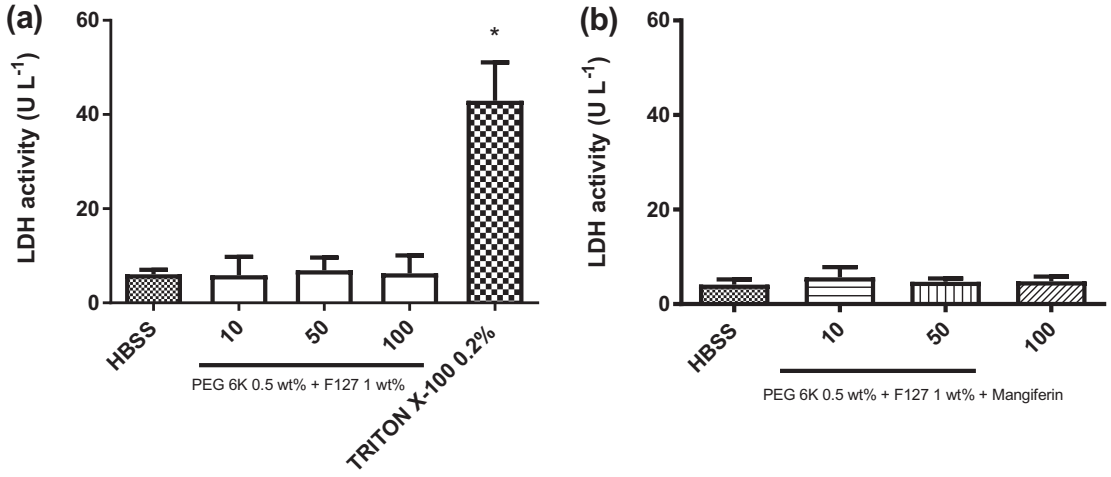


Fig. 2. Cytotoxicity of PEG/F127 system: (a) absence or (b) presence of mangiferin, measured by the lactate dehydrogenase (LDH) assay. Triton X-100 is a standard cytotoxic agent. (Data represent mean ± SEM. * = $p < 0.05$; ANOVA, Tukey as *post hoc test*.)

PEG 35 K. The addition of at least 5 wt% of PEG 35 K seems to be efficient to increase approximately ~3.3× (S-S_0/S_0) mangiferin water solubility, and more (~4.17x) if double amount of PEG is added. However, in the case of 10 wt% of PEG 6K, it is observed a solubility increase of ~3.08×. Indeed, the PEG addition evaluation in F127 diluted solutions was previously reported by our work's group using a model drug griseofulvin, where PEG 400, 6 K and 35 K added at different concentrations only show an efficient drug solubility for PEG 35 k in the concentration range of 0.5–2 wt% [6].

In Fig. 1b, it can be noticed the nearly linear solubilisation capacity profile, increasing PEG concentration range. The experimental results confirm the efficiency by the addition of a minimum amount of PEG in 1 wt% F127, once all S-S_0/$wt\%$ values are higher than F127 pure solution. This confirms the efficiency of PEG addition at F127 diluted solutions, considering the solubilisation of hydrophobic drugs in aqueous medium, which may decrease the over-usage of surfactants in pharmaceutical formulations. For instance, in this work, the addition of just 0.5 wt% PEG 6 K at1 wt % F127 showed a higher mangiferin solubilisation value (7.92 mg dL^{-1}) when compared to F127 1 wt% pure solution (2.05 mg dL^{-1}).

Table 1 shows the results of α-glucosidase inhibition and IC_{50}-values of the free-form mangiferin and mangiferin-loaded PEG 6 K 0.5wt%/F127 1wt%micellar system.

The IC_{50} value for mangiferin-loaded PEG/F127 is much lower than the free-form drug, (Table 1), i.e. the drug loading system greatly affected the IC_{50} of mangiferin, showing a synergistic effect between the system and the bioactive. Although the action mechanism of mangiferin inhibition is not widely known [3], our results for mangiferin/PEG/F127 DDS showed a high value of α-glucosidase inhibitory activity, 95.42 ± 3.75, which can be a potential alternative DDS in the treatment of diabetes.

The same system was then evaluated regarding its cytotoxicity, as shown in Fig. 2a and b.

For human neutrophils, both tested unloaded (Fig. 2a) and mangiferin-loaded systems (Fig. 2b) did not significantly increase the LDH activity in cell suspension, showing no cytotoxicity in comparison to Triton-X 100 at 0.2%.

4. Conclusions

PEG/F127 micellar system greatly contributes to a better environment for mangiferin. Additionally, *in vitro* tests showed outstanding results, because the analyzed mangiferin-loaded PEG/F127 system showed no cytotoxic effect, with low % LDH activity, and also increased the % α-glucosidase inhibitory activity of the antidiabetic mangiferin. Therefore, PEG addition to F127 micellar solutions seems to be a smart DDS strategy to increase the biodisponibility of hydrophobic drugs, because it might decrease the over-usage of surfactants and enhance mangiferin α-glucosidase inhibition.

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